

data: inputs x , target y

stage 1: $F_0 = \text{mean of } y$

pseudo-residuals
 $r = y - \text{current prediction}$

next stage

fit a regression tree h
inputs x , target r

add it, scaled:
 $F \leftarrow F + \eta \cdot h(x)$

after M trees

final model
 $F_0 + \eta h_1 + \eta h_2 + \cdots + \eta h_M$